

# David G Courtney BSc PhD

Senior Lecturer in Molecular Virology, Queen's University Belfast  
UK Ambassador for the European Research Council

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Molecular virologist and RNA biologist. David leads The Courtney Group at Queen's University Belfast, which studies how influenza A virus RNA recruits host machinery to be modified, exported, translated and trafficked, and how those dependencies can be turned into RNA-based therapeutics. He has won over €1.8M in competitive funding as sole PI, sits on BBSRC and European funding panels, and represents the United Kingdom as an Ambassador for the European Research Council.

1,531 citations · h-index 19 · 52 indexed works (OpenAlex, 2026-10-07)

## APPOINTMENTS

|             |                                                                                                                                                                                           |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025 –      | <b>Senior Lecturer in Molecular Virology</b> , Queen's University Belfast, UK                                                                                                             |
| 2021 – 2025 | <b>Lecturer and ERC Research Fellow</b> , Queen's University Belfast, UK<br>Founded The Courtney Group with an ERC Starting Grant.                                                        |
| 2020 – 2021 | <b>Postdoctoral Research Fellow in Molecular Virology</b> , Queen's University Belfast, UK<br>High-throughput drug repurposing screens against SARS-CoV-2 in airway epithelial cells.     |
| 2019 – 2020 | <b>Marie Skłodowska-Curie Postdoctoral Fellow</b> , Institut Pasteur, Paris, France<br>Developed methods to identify the host RNA-binding proteins of influenza A virus RNAs.             |
| 2016 – 2019 | <b>Marie Skłodowska-Curie Postdoctoral Fellow</b> , Duke University, North Carolina, USA<br>Showed how m6A and m5C RNA modifications regulate influenza A virus, HIV-1 and other viruses. |
| 2010 – 2011 | <b>Industrial placement, BioPharm R&amp;D</b> , GlaxoSmithKline, UK<br>Twelve months in molecular expression and cell line development.                                                   |

## EDUCATION

|             |                                                                                                                                                             |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2012 – 2015 | <b>PhD, Molecular Biology</b> , Ulster University, UK<br>The development of personalised medicines for corneal dystrophies (siRNA and CRISPR therapeutics). |
| 2008 – 2012 | <b>BSc (Hons) Biomedical Sciences with industrial studies, First Class</b> , Ulster University, UK                                                          |

## LEADERSHIP AND SERVICE

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|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025 –      | <b>UK Ambassador for the European Research Council</b> , Ambassadors for the ERC network<br>One of 32 ERC grantees chosen to make the case for frontier research to policymakers, the media and the research community.         |
| Current     | <b>Vice-Chair, Marie Skłodowska-Curie Actions Postdoctoral Fellowships panel</b> , European Commission                                                                                                                          |
| 2024 –      | <b>Member, BBSRC Panel of Experts</b> , UK Research and Innovation                                                                                                                                                              |
| Current     | <b>Elected member, Virology Division Committee</b> , Microbiology Society                                                                                                                                                       |
| 2024        | <b>Chair and organiser, UK ERC Showcase</b> , Queen's University Belfast<br>Flagship event with panels featuring Nobel Laureate Prof Emmanuelle Charpentier and Prof Teresa Lambe OBE.                                          |
| 2023 – 2024 | <b>Contributor, US National Academies report “Toward Sequencing and Mapping of RNA Modifications”</b> , US National Academy of Sciences<br>Invited ideation panel delegate (2023) and co-author of a commissioned paper (2024). |
| 2021 – 2024 | <b>Grant reviewer</b> , MSCA (reviewer and rapporteur), US National Science Foundation panel, Swiss National Science Foundation, Polish National Science Centre                                                                 |
| Current     | <b>Module Coordinator, MSc Research Project</b> , Wellcome-Wolfson Institute for Experimental Medicine, Queen's                                                                                                                 |
| Current     | <b>STEM NI Ambassador</b> , Northern Ireland                                                                                                                                                                                    |
| 2014 –      | <b>Fellow of the Winston Churchill Memorial Trust</b> , Churchill Fellowship, UK                                                                                                                                                |

## GRANTS AND AWARDS

|             |                                                                                                                                                                                               |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2021 – 2026 | <b>ERC Starting Grant: PTFU, post-transcriptional regulation of influenza A virus RNA</b> , European Research Council · sole PI · grant 948834<br>€1,568,010 to establish The Courtney Group. |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

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|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2023 – 2026 | <b>MRC research grant: host ATPases in rhinovirus replication</b> , UKRI Medical Research Council · co-investigator · MR/X020371/1<br>£458,574.                                                                             |
| 2024 – 2025 | <b>Royal Society Partnership Grant</b> , The Royal Society · with Regent House School<br>A research project run with secondary school pupils.                                                                               |
| 2017 – 2020 | <b>Marie Skłodowska-Curie Global Fellowship: IAV-m6A</b> , European Commission · sole PI · Duke University and Institut Pasteur<br>€260,000 to study m6A RNA methylation in influenza A virus replication and pathogenesis. |
| 2015        | <b>Postgraduate Student of the Year (John RE Scott Convocation award)</b> , Ulster University                                                                                                                               |
| 2014 – 2015 | <b>Churchill Fellowship</b> , Winston Churchill Memorial Trust<br>€7,600 for research stays in Italy and Denmark.                                                                                                           |

## PATENT APPLICATIONS

|      |                                                                                                                                                                                                                      |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024 | <b>Novel compounds for the treatment of viral infections</b> , US patent application 18/550,005                                                                                                                      |
| 2019 | <b>Compositions and methods for enhanced gene expression and viral replication</b> , US patent application 16/091,822<br>Using m6A-associated cellular machinery to boost transgene expression or viral replication. |
| 2019 | <b>Single guide RNA/CRISPR/Cas9 systems, and methods of use thereof</b> , US patent application 16/326,908<br>Allele-specific CRISPR/Cas9 systems for treating corneal dystrophies.                                  |

## SELECTED TALKS AND EVENTS

|      |                                                                                                    |
|------|----------------------------------------------------------------------------------------------------|
| 2026 | <b>Invited speaker: Building ERC-competitive research</b> , University of Split, Croatia           |
| 2024 | <b>Event chair: UK ERC Showcase</b> , Queen's University Belfast                                   |
| 2023 | <b>Delegate: ideation panel on RNA modification mapping</b> , US National Academy of Sciences      |
| 2020 | <b>Invited speaker: ANRS International Symposium on RNA Modifications</b> , Institut Cochin, Paris |

## MENTORING AND LEADERSHIP

- Mentored Dr Hannah Turkington to a competitive three-year BBSRC Postdoctoral Fellowship (2024).
- PhD graduates went straight to postdoctoral posts at the University of Cambridge and the University of Copenhagen.
- MSc graduates have moved into government science roles, including Forensic Science NI.
- EMBO Laboratory Leadership course and UKRI Future Leaders Fellows leadership retreat (2024).

## PUBLIC ENGAGEMENT AND MEDIA

- Regular expert voice on BBC television and national radio during the COVID-19 pandemic, explaining viral mutation, immune escape and variant surveillance.
- Long-form articles for The Conversation reaching international readers.
- Works with regional health bodies and schools to promote influenza vaccination awareness.

## PROFESSIONAL MEMBERSHIPS

Microbiology Society · American Society for Virology · RNA Society

## SELECTED PUBLICATIONS

# co-senior author · \* co-first author. Full list at [davidgcourtney.com/publications](https://davidgcourtney.com/publications) and on ORCID.

1. Bonazza S, Turkington HL, Sukumar S, Peate E, Coutts HL, Montgomery JJ, Hawthorn C, Getty EMPE, Touzelet O, Barabas J, Power UF, **Courtney DG** (2026). Myoferlin is a component of late-stage vRNP trafficking vesicles for enveloped RNA viruses. *Nature Communications* 17. Cited 1.
2. Bonazza S, **Courtney DG** (2025). Influenza A virus RNA localisation and the interceding trafficking pathways of the host cell. *PLoS Pathogens* 21(4): e1013090. Cited 10.
3. Bonazza S, Coutts HL, Sukumar S, Turkington HL, **Courtney DG** (2024). Identifying cellular RNA-binding proteins during infection uncovers a role for MKRN2 in influenza mRNA trafficking. *PLoS Pathogens* 20(5): e1012231. Cited 11.
4. Dupont M, Krischuns T, Giai-Gianetto Q, Paisant S, Bonazza S, Brault J, Douché T, Perez-Perri JI, Hentze MW, Cusack S, Matondo M, Isel C, **Courtney DG**#, Naffakh N# (2024). The RBPome of influenza A virus mRNA reveals a role for TDP-43 in viral replication. *Nucleic Acids Research*. Cited 18.

5. Coutts LH, **Courtney DG** (2024). Identifying individual pseudouridine (Ψ) sites across transcripts from HIV-1 infected cells. *HIV Protocols. Methods in Molecular Biology*, vol 2807.
6. **Courtney DG**, Tsai K, Bogerd HP, et al. (2019). Epitranscriptomic regulation of HIV-1 gene expression by m5C. *Cell Host & Microbe* 26(2): 217–227. Cited 261.
7. **Courtney DG**, Chalem A, Bogerd HP, et al. (2019). Extensive epitranscriptomic methylation of A and C residues on murine leukemia virus transcripts enhances viral gene expression. *mBio* 10(3): e01209-19. Cited 79.
8. **Courtney DG**, Kennedy EM, Dumm RE, et al. (2017). Epitranscriptomic enhancement of influenza A virus gene expression and replication. *Cell Host & Microbe* 22(3): 377–386. Cited 250.
9. **Courtney DG**, Moore JE, Atkinson SD, et al. (2015). CRISPR/Cas9 DNA cleavage at SNP-derived PAM enables both in vitro and in vivo KRT12 mutation-specific targeting. *Gene Therapy* 23(1): 108–112. Cited 129.
10. **Courtney DG**, Toftgaard Poulsen E, Kennedy S, et al. (2015). Protein composition of TGFBI-R124C- and TGFBI-R555W-associated aggregates suggests multiple mechanisms leading to lattice and granular corneal dystrophy. *Investigative Ophthalmology & Visual Science* 56: 4653–4661. Cited 39.

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